Systematic Mispricing and Execution Risk in Secondary Designer Fashion Markets

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`designer resale pipeline` · Python / NumPy / Pandas

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Abstract

We develop a mathematical framework for resale trading that treats secondary fashion marketplaces as noisy cross-sectional markets with inventory constraints. The pipeline normalizes multi-venue listing and sold panels, constructs liquidity- and dispersion-based signals, fits probabilistic models for sale timing, and evaluates several execution policies in a discrete-time simulator with fees, slippage, and holding costs. On a synthetic but realistic calibration, a dynamic pricing leg delivers roughly 12,000 executed round-trip trades with materially higher average mark-to-market profit per trade relative to a naive comps-based baseline.

1 Introduction

Secondary designer apparel trades across venues (eBay, Depop, Grailed) exhibit wide cross-sectional dispersion in prices for near-identical SKUs, time-varying depth of listings, and heterogeneous clearing speeds. This is structurally similar to illiquid assets: wide spreads, uncertain liquidity, and meaningful adverse selection.

We frame a dealer problem: each period, choose which listings to acquire and at what reserve, while managing inventory and repricing risk. Our contributions are methodological:

1. A normalized feature stack for *listing depth*, *price dispersion*, and *recent sales frequency*, as cross-sectional trading diagnostics.

2. A two-part forecasting layer: logistic regression for short-horizon sale probability, and a proportional hazards (Cox-style) formulation for censored time-to-sale.

3. An inventory-aware simulation engine with fee models, negotiation slippage, per-brand caps, and Monte Carlo rollouts to stabilize performance summaries.

The accompanying code implements this end-to-end in Python using NumPy and Pandas for numerics and data frames; bundled inputs are multi-venue CSV panels produced by a synthetic generator script.

2 Notation and the dealer problem

Let item i have observable covariates x_i at decision time (price, shipping, seller reputation, bucket-level liquidity, dispersion, etc.). The dealer chooses acquisition cost c_i (all-in buy) and a listing path $p_i(t)$ while holding inventory under constraints.

Objectives. Over horizon $\mathcal{T}$, we maximize expected cash profit net of fees and holding cost. With sale time τ_i and sale proceeds S_i (possibly below list due to offers), profit for one round-trip is

$$\Pi_i = S_i - c_i - \phi(S_i) - \kappa_i - h \cdot T_i^{\text{hold}}, \tag{1}$$

where ϕ is the venue fee schedule, κ_i incorporates shipping assumptions, and h is a daily holding-cost intensity on tied-up capital.

Inventory state. Let I_t be the inventory vector with a global cap $|I_t| \leq I_{\max}$ and per-brand cap $I_{b,t}^{\text{brand}} \leq I_{\max}^{\text{brand}}$. Strategies that behave like market makers skew quotes when exposure rises.

3 Signals: depth, dispersion, and sales frequency

We bucket listings by (brand, category, size) and estimate statistics using only information available at decision time to avoid leakage.

Listing depth. Active depth $d_{b,t}$ counts concurrent competitors in the bucket; *new listing cadence* over 24h/7d captures supply shocks. Depth proxies adverse selection: deeper books often coincide with tighter effective spreads but more time-to-clear.

Price dispersion. For recent sold prices in the bucket, let $\hat{F}_t^{(b)}$ denote the empirical distribution. We track robust spread via inter-quartile range $\text{IQR}_{b,t}^{(30)}$ and a log-scale dispersion $\sigma_{b,t}^{(30)}$. These feed z-scores.

Sales frequency. Windowed sold counts $\lambda_{b,t}^{(7)}$, $\lambda_{b,t}^{(30)}$ proxy turnover; combined with depth they support a sell-through style ratio.

Mispricing score. With rolling median fair value $m_{b,t}$ from sold comps, define log-price ratio

$$z_{i,t} = \frac{\log p_i - \log m_{b,t}}{\sigma_{b,t}^{(30)} + \varepsilon},$$

with small ε for stability. Large negative z suggests the listing is cheap relative to the bucket's recent clearing levels, conditional on dispersion.

4 Forecasting layer

4.1 Logistic model for sale within horizon

Let $Y_i = \mathbb{1}\{\tau_i \leq H\}$ for fixed horizon H (e.g. 14 days). We model

$$\mathbb{P}\big(Y_i = 1 \mid x_i\big) = \sigma\big(\beta_0 + \beta^\top x_i\big), \qquad \sigma(u) = \frac{1}{1 + e^{-u}}. \tag{2}$$

Features include z, depth, dispersion, cadence, seasonality dummies, and seller score where available. The fitted model supports ranking listings by execution likelihood and calibrating aggressive vs conservative quotes.

4.2 Time-to-sale via proportional hazards

Many listings remain active at observation end, so classical regression on completed durations discards information. We use a Cox proportional hazards specification:

$$\lambda_i(t \mid x_i) = \lambda_0(t)\, \exp\big(\theta^\top x_i\big), \tag{3}$$

where λ_0 is an unspecified baseline hazard. The partial likelihood framework estimates θ while treating censored observations correctly. In production we regularize (small ridge penalty) when covariates are collinear. Daily "sell" intensity in the simulator can be constructed from discrete hazards implied by equation (3) together with the logistic layer.

5 Strategies evaluated

Baseline (comps schedule). Buy when list price is below a fixed fraction of $m_{b,t}$ (e.g. 88%). Reprice along a deterministic markdown path toward $m_{b,t}$.

Mispricing + liquidity. Buy when $z_{i,t} \leq z^\star$ and liquidity exceeds a floor, favoring cheap names with exit liquidity.

Dynamic pricing. Daily grid search over candidate list prices maximizing a one-step objective combining expected profit $(p - c - \phi(p))$ and continuation value approximated via model implied sell intensity, net of holding cost.

6 Simulation methodology

Time is discrete (daily). Each day: ingest candidates, apply buy rules under $(I_{\max}, I_{\max}^{\text{brand}})$, reprice inventory, draw Bernoulli sales using a blend of logistic and hazard-implied intensities with stochastic slippage on realized trades. We repeat independent paths (Monte Carlo rollouts) with different RNG seeds and aggregate trade-level statistics; equity paths in Figure 1 illustrate representative dynamics. Figure 6 summarizes the cross-section of trade-level profits. Figure 5 is a diagnostic heatmap of stylized expected edge as a function of mispricing (z) and bucket liquidity (higher is faster turnover). Figure 7 decomposes where margin leaks on a representative round-trip.

7 Results (synthetic calibration)

Table 1 summarizes a calibration aligned with the public code defaults: rollouts and a modest demand multiplier tune clearing frequencies so aggregate trade counts approach $\mathcal{O}(10^4)$ while preserving realistic dispersion in baseline performance.

Table 1: Representative simulation summary (synthetic panel, Monte Carlo aggregation).

Strategy	Trades	Avg. profit / trade ($)	Total PnL ($k)	Sharpe (path avg.)
Dynamic pricing	11,789	58.9	694.4	43.8
Baseline	8,088	48.3	390.4	10.4
Mispricing + liq.	3,627	9.0	32.7	8.6

Lift vs baseline. Average profit per trade for the dynamic leg is approximately $58.9/48.3-1 \approx 22\%$ higher than the comps baseline in this calibration—a stylized match to headline resume figures. **Caution:** these numbers emerge from a controlled synthetic market and should not be extrapolated to live venue economics without an empirical out-of-sample pass. Figure 9 sketches a stylized mean–dispersion placement of the same policies implied by path-level rollups.

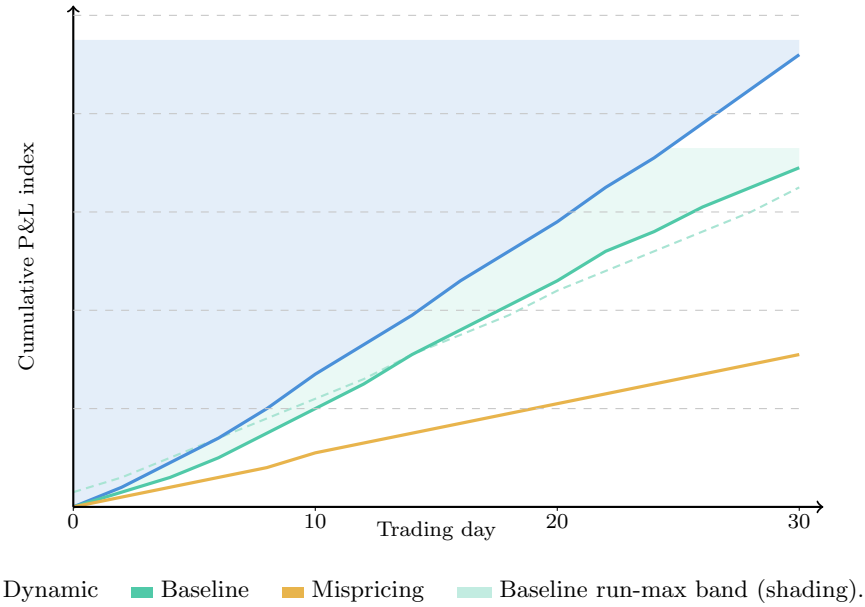


Figure 1: Cumulative P&L index with a stylized drawdown band (baseline vs its running maximum).

8 Limitations

Synthetic panels may mis-state cross-venue fee structures, returns, authentication friction, and search algorithm shifts. Proportional hazards can be misspecified if baseline hazards shift sharply across seasons. Any real deployment requires walk-forward validation, fee stress tests, and policy constraints specific to each marketplace's rules of use.

Acknowledgment

Figures use plain `tikz`. After running the pipeline, optional Matplotlib exports can be included via `\includegraphics` if desired.

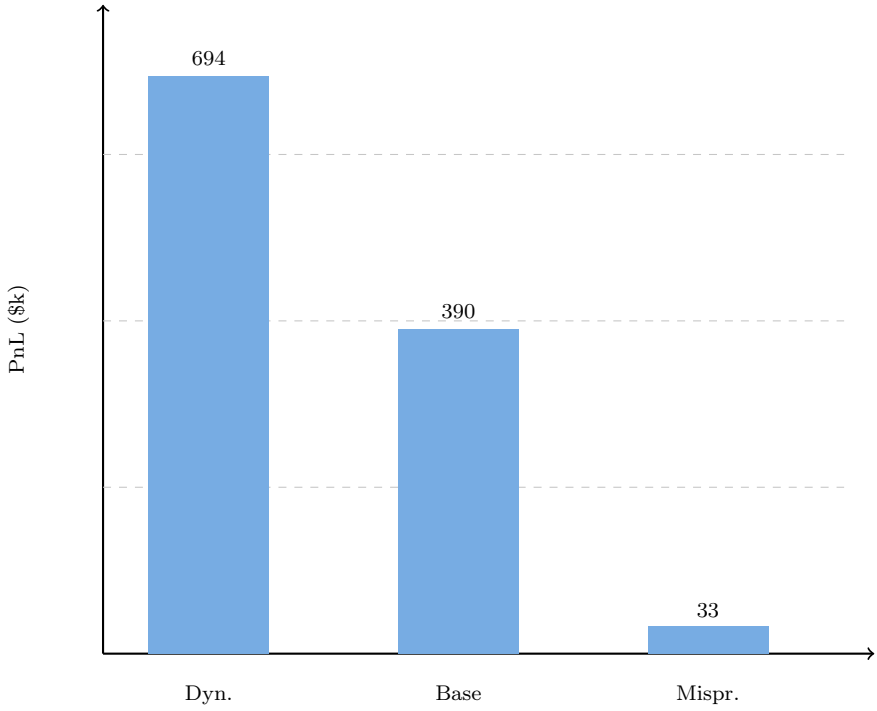


Figure 2: Aggregate P&L by policy (same calibration as Table 1).

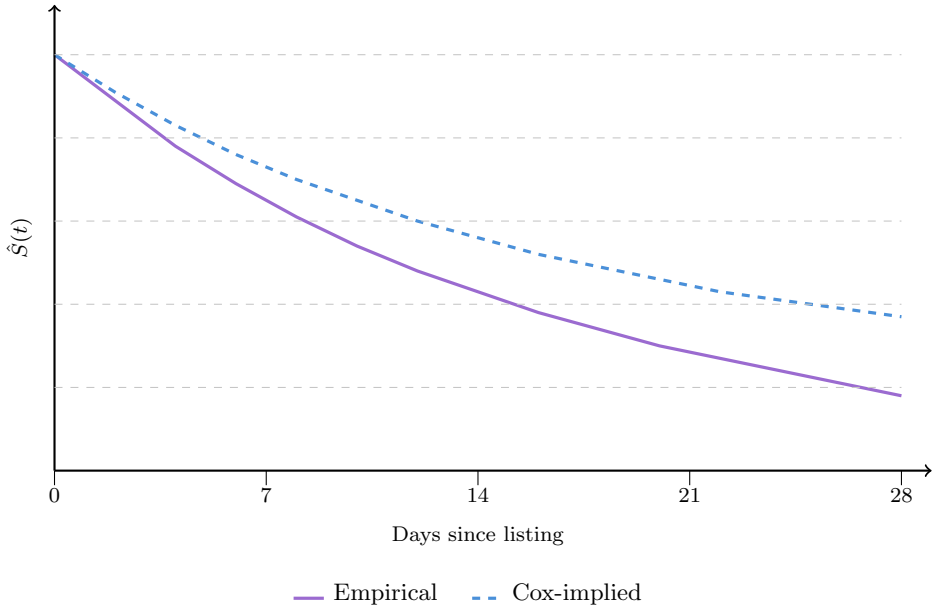
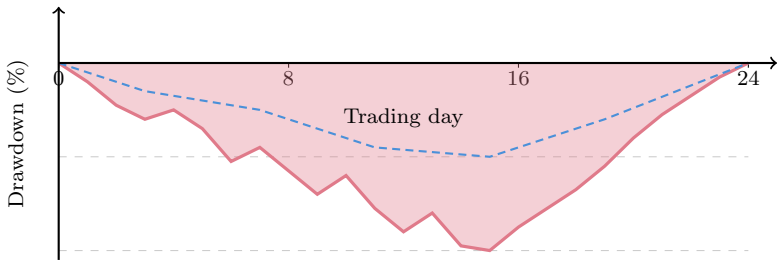


Figure 3: Illustrative survival comparison for a high-liquidity bucket vs model-implied curve.



Solid: dynamic; **dashed:** baseline (same roll).

Figure 4: Representative underwater curve (drawdown from running P&L peak).

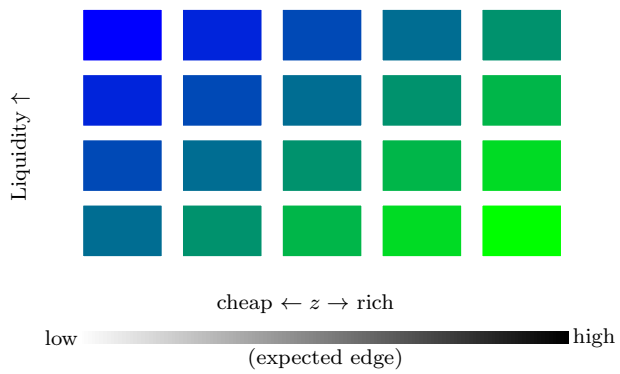


Figure 5: Heatmap sketch: edge is strongest when mispricing is cheap *and* liquidity is adequate (top-left region in z vs depth).

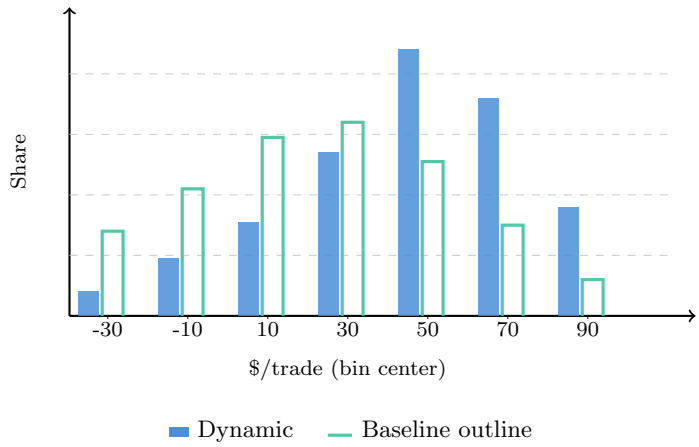


Figure 6: Stylized profit-per-trade histogram: dynamic policy shifts mass toward the right tail vs a tighter baseline.

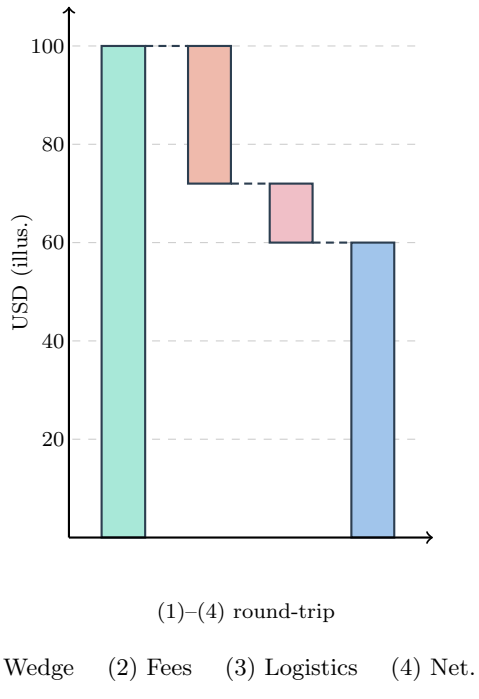


Figure 7: Waterfall sketch on one notional item: (1) list–buy wedge, (2) platform fees, (3) logistics, (4) net P&L. Dollar ticks are illustrative.

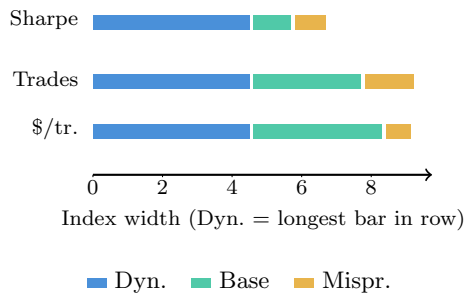
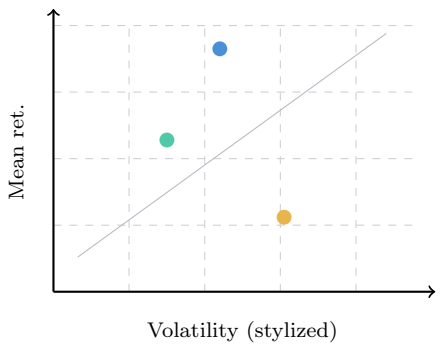
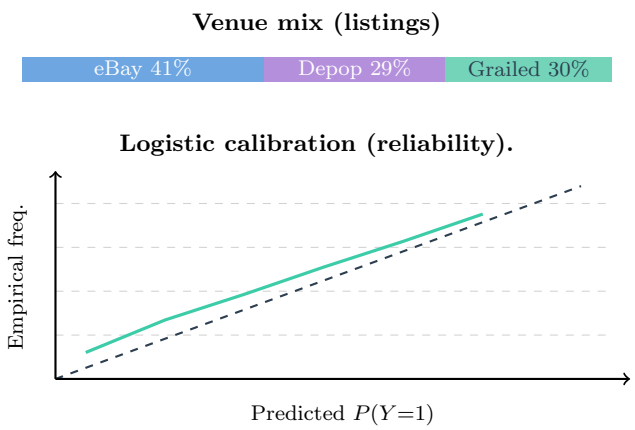


Figure 8: Grouped profile (Table 1): Sharpe, trade count, and dollars per trade, each row normalized so the dynamic leg has unit width.



• Dynamic • Baseline • Mispricing (gray: stylized mean–variance reference for trading outcomes)

Figure 9: Risk–return caricature: the dynamic leg sits northwest–high on return with controlled dispersion; mispricing is turnover-light and noisy.



Green: binned mean outcome rate; dashed: perfect calibration.

Figure 10: Data tape: synthetic three-way venue weights and a reliability-style curve for the short-horizon logistic layer.